# vllm-sr/d3-lite

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## d3-lite

**d3-lite** is the 0.8B multimodal foundation decision model of Decision 3.0, the decision models of vLLM Semantic Router. Give it an input (text or JSON, optionally with images and videos) and the questions you need answered: pick one of several options, say yes or no, or rate on a scale. It answers them all in one call and returns a probability for every answer, without generating text.

| | |
|---|---|
| **Parameters** | 0.85B, including the 0.10B vision encoder |

Downloads last month
879

Safetensors
Model size 0.9B params
Tensor type BF16 · Chat template
Files info

Inference Providers NEW
Zero-Shot Classification
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Model tree for vllm-sr/d3-lite
Base model Qwen/Qwen3.5-0.8B-Base
Finetuned Qwen/Qwen3.5-0.8B
Finetuned (518) this model
Finetunes 2 models
Quantizations 8 models

| | |
|---|---|
| **Inputs** | Text or JSON, plus images and videos (several per request) |
| **Decision types** | Choice · Yes / No · Score |
| **License** | Apache-2.0 |

## Highlights

- **Jev Decision Index 0.3.2: 32.3 on the official text board**, first among the entries of similar size and on the board's Pareto frontier; public suite 36.93, measured with the official 0.3 kit on the released weights: all 140,178 public requests answered, none unsupported.
- **+14.9 over Decision 2.0** (its 0.8B model: 17.5 on the official board): +16.4 on the public suite, ahead in all five of its areas, and +13.5 and +11.8 on the private same-skill tests and new-domain tasks.
- **Reads images:** multiple images per request (PNG, JPEG or WebP), given as paths, URLs, PIL images or base64 data URLs; every question of the request sees all of them.
- **Reads videos:** multiple videos per request (MP4, WebM, MOV or MKV), given as paths, URLs, base64 data URLs or frame arrays, read at 2 frames per second; **Perception Test (multiple-choice video QA): 59.3** (internal evaluation).
- **Speed:** a median of 13.6 ms for a text request, 64.0 ms for a request with an image and 291 ms for a request with a 10-second video, on one

**Spaces using** `vllm-sr/d3-lite` 3

- vllm-sr/d3-lite-decision-model
- multimodalart/jev-decision-index
- tardellirs/model-pulse

**Collection including** `vllm-sr/d3-l…`

**Decision 3.0** Collection
Towards O… • 7 items • Upd… • △ 97

AMD Instinct MI325X GPU, one request at a time with AutoModel.

- **Serving:** 518 requests/s with p99 under 1 s on one AMD Instinct MI325X with vLLM Semantic Router, 7.1× AutoModel (one-question text, `batching` profile); the default `exact` profile gives answers identical to AutoModel's (Serving performance).
- **Many questions, one call:** Choice, Yes / No and Score questions about the same input are answered together, each from its own forward pass over the input, with a probability for every option.

## Quickstart

```
pip install "transformers==5.17.0" torch tor
pip install flash-linear-attention  # option
```

```
import json

from huggingface_hub import hf_hub_download
from transformers import AutoModel

model = AutoModel.from_pretrained("vllm-sr/d

# Text
result = model.system_one(
    state="The order arrived damaged yesterd
    questions={
        "route": {
            "type": "choice",
            "instructions": "Which team shou
            "criteria": {
                "returns": "Refunds, replace
                "billing": "Payments, invoic
                "technical": "Product setup
```

```
            }
        },
        "receipt": {
            "type": "noul",
            "instructions": "Does the custom
        },
        "urgency": {
            "type": "score",
            "instructions": "How urgent is t
            "criteria": [
                "Routine",
                "Soon",
                "Today"
            ]
        }
    },
)
print(json.dumps(result["answers"], indent=2

# Text and an image (or several)
receipt = hf_hub_download("vllm-sr/d3-lite",
result = model.system_one(
    state="The customer says the blender arr
    images=[receipt],  # local paths, http(s
    questions={
        "route": {
            "type": "choice",
            "instructions": "Which team shou
            "criteria": {
                "returns": "Refunds, replace
                "billing": "Payments, invoic
                "technical": "Product setup
            }
        },
        "on_receipt": {
            "type": "noul",
            "instructions": "Does the receip
        },
        "payment": {
            "type": "choice",
            "instructions": "How was the ord
```

```
            "criteria": {
                "card": None,
                "cash": None,
                "gift card": None
            }
        }
    },
)
print(json.dumps(result["answers"], indent=2

# Text and a video (or several)
clip = hf_hub_download("vllm-sr/d3-lite", "as
result = model.system_one(
    state="A short clip from a test camera."
    videos=[clip],  # local paths, http(s) U
    questions={
        "direction": {
            "type": "choice",
            "instructions": "Which way does
            "criteria": {
                "right": "From left to right
                "left": "From right to left"
                "still": "It does not move"
            }
        },
        "color_change": {
            "type": "noul",
            "instructions": "Does the square
        }
    },
)
print(json.dumps(result["answers"], indent=2

# Or as a pipeline:
# transformers.pipeline("decision", model="v
```

Images go before the text of the request, each read at up to 1.6 megapixels; videos follow the images, read at 2 frames per second (at most 32 frames spread over the whole video, each at up to 0.2

megapixels). Every question of the request sees all of them.

### Run with vLLM Semantic Router

vLLM Semantic Router serves Decision 3.0 natively in Engine mode, which runs one model behind the System One API on your own machine. Its answers match this package's own `d3_server.py` exactly, for text, images and videos (measured on AMD Instinct MI325X).

```
curl -fsSL https://vllm-sr.ai/install.sh | ba
export PATH="$HOME/.local/bin:$PATH"
vllm-sr serve -e vllm-sr/d3-lite
```

The platform is detected automatically; choose one with `--platform auto|cpu|cuda|rocm` (default `auto`).

Engine mode and Decision 3.0 are in the development channel (`--channel dev`) until a stable vLLM Semantic Router release includes them. The first start pulls the image for your machine and downloads the model. Then ask questions about a request:

```
curl -s localhost:8899/v1/systemone -H 'conte
  "model": "vllm-sr/d3-lite",
  "state": "Write a Python function that merg
  "questions": {
    "kind": {"type": "choice", "instructions
             "criteria": {"code": "Writing o
    "reasoning": {"type": "noul", "instructi
  }
}'
```

Send images as base64 data URLs (PNG, JPEG or WebP) in `images` and videos (MP4, WebM, QuickTime or Matroska) in `videos`; every question sees them:

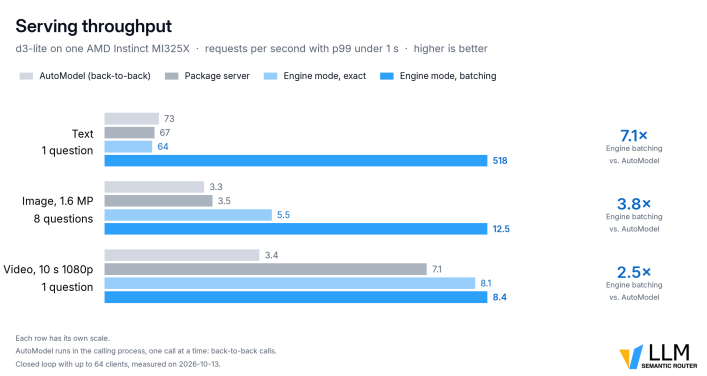
```
curl -s localhost:8899/v1/systemone -H 'conte
{
  "model": "vllm-sr/d3-lite",
  "state": "What does the attached image sho
  "images": ["$(base64
  "questions": {
    "kind": {"type": "choice", "instructions
             "criteria": {"chart": "A chart
  }
}
EOF


curl -s localhost:8899/v1/systemone -H 'conte
{
  "model": "vllm-sr/d3-lite",
  "state": "What happens in the clip?",
  "videos": ["data:video/mp4; base64,$(base64
  "questions": {
    "moving": {"type": "noul", "instructions
  }
}
EOF
```

`vllm-sr stop` stops it. See Router and Engine modes for limits and options.

### Serving performance

**On one AMD Instinct MI325X, vLLM Semantic Router serves d3-lite at 518 requests/s with p99 under 1 s, 7.1× AutoModel** (one-question text, `batching` profile), and its default `exact` profile gives answers identical to AutoModel's.




AutoModel runs in your own process, one call at a time (the Quickstart code); the package server is its `python d3_server.py`; Engine mode is `vllm-sr serve -e` with its default `exact` profile, or with `--runtime-profile batching`.

Where Engine mode helps, and what it costs:

- **Throughput:** 7.1× AutoModel's requests per second on one-question text, 3.8× on 8-question images and 2.5× on 10-second videos, with the better Engine profile of each row.
- **Start-up:** the first answer comes 12.4 s after launch with empty kernel caches, against 25.0 s for AutoModel, whose first image then takes another 1.15 s, and 37.5 s for the package server.
- **Many questions and long inputs:** one request at a time, Engine mode (`exact`) answers 16 text questions in 38.0 ms (AutoModel: 38.2 ms), 8 questions about one image in 238 ms (305 ms) and a 32k-token text in 359 ms (381 ms); `batching` runs the 8 image questions together in 133 ms, 56 % less than AutoModel.
- **GPU memory:** 16.1 GiB per replica at peak with `exact`, above the package server's 12.6 GiB, and 16.3 GiB with `batching`; AutoModel, serving one request at a time, peaked at 5.7 GiB. Engine

mode keeps up to 2 GiB of image features cached.

- **Answers:** Engine mode (`exact`) returned answers byte-identical to AutoModel's on all 7 request types (text with 1, 4 or 16 questions, an image with 1 or 8, the video and a 32k-token text) when AutoModel used the same kernel settings: 140 Engine replies and 21 AutoModel calls compared. With the kernel autotuning AutoModel runs in each new process, all 7 were byte-identical too. `batching` changed 0.172 % of decisions on the complete public suite (555 of 323,259 questions, near-ties), with gold accuracy 57.994 → 57.997 % and the index 36.86 → 36.87.

A single short request is fastest in your own process: AutoModel answers one text question 2.6 ms sooner than Engine mode and one new image 32 ms sooner, the cost of going through the router to the model worker and its media path. Engine mode's gains come from serving many requests at once.

▶ Single-request latency and the numbers behind the chart

Measured on 2026-10-13, one AMD Instinct MI325X per replica and each side alone on the GPU: ROCm 7.2 with PyTorch 2.12; AutoModel and the package server with Transformers 5.17.0; vLLM Semantic Router Engine mode built from source, with serving optimizations that are newer than its published images. Requests: a support ticket of about 330 tokens with 1, 4 or 16 questions; a 1,456 × 1,092 PNG (1.6 MP) with 1 or 8 questions; a 10-second 1080p MP4, read at 2 frames per second; a 32k-token text; every image and video request carries new bytes. One request at a time: 3 warm-up calls, then up to 20 timed calls. Under load: 1, 2, 4, … 64 closed-loop clients, each level

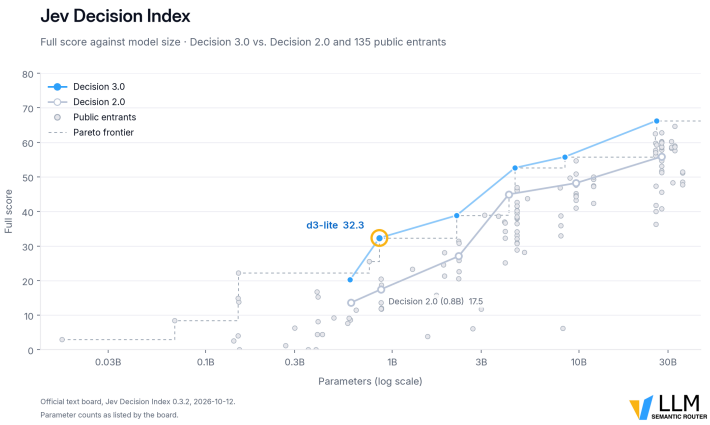
warmed up and then counted for 15 to 60 s; Engine mode admits 32 requests in flight per worker and answers the rest at once with 503. Start-up: model files on local disk; Engine mode from the start of its router process, without the CLI's image pull.

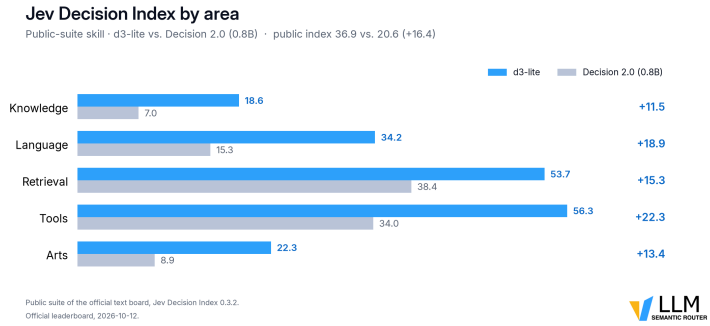
## Evaluation

### Text: Jev Decision Index 0.3.2

| Model | Jev Decision Index ↑ | Public ↑ | Same-skill tests ↑ | New-domain tasks ↑ |
|---|---|---|---|---|
| **d3-lite** | **32.3** | **36.9** | **34.8** | **25.7** |
| jiwo 0.8B v2 | 25.6 | 29.4 | 30.6 | 17.9 |
| OneJev 0.8B | 20.5 | 22.2 | 21.8 | 19.9 |
| JPT-0.8B | 18.7 | 19.4 | 17.8 | 22.0 |
| Sifr 0.8B v3.1 | 17.8 | 25.7 | 18.9 | 15.2 |
| Decision 2.0 (0.8B) | 17.5 | 20.6 | 21.3 | 13.9 |

On the official text board, d3-lite is on the Pareto frontier, where no entry with as many parameters or fewer scores higher, and ranks first among the ten entries within 0.75 to 1.33 times its size. Five of the six Decision 3.0 models are on that frontier; d3-edge, second among the entries of its size, is not.




## Images: Jev Decision Index vision board 0.3.2

| Model | Vision Index ↑ | Public ↑ | Private ↑ |
|---|---|---|---|
| **d3-lite** | **42.1** | **47.9** | **36.3** |
| JPT-0.8B | 42.1 | 44.6 | 39.7 |
| OneJev 0.8B | 41.5 | 46.5 | 36.5 |
| Intern-Decision-0.8B | 36.3 | 39.0 | 33.7 |

On the official vision board, the six Decision 3.0 models form the whole Pareto frontier: no other entry is on it. d3-lite ranks first among the four entries within 0.75 to 1.33 times its size, level with JPT-0.8B at 42.1.

d3-lite on the public vision benchmarks († approximate rebuild):

| Benchmark | d3-lite |
|---|---|
| CV-Bench | 65.5 |
| BLINK | 37.7 |
| RealWorldQA | 35.9 |
| CharXiv † | 65.7 |
| InfographicVQA † | 74.9 |
| Mind2Web † | 53.1 |
| Winoground | 57.0 |
| KIE (CORD+FUNSD) † | 93.2 |
| Moderation (Hateful Memes) | 11.8 |
| R-Bench-M | 10.5 |
| MMMU-Pro vision | 8.3 |

### Videos: Perception Test

| Perception Test (validation) | d3-lite |
|---|---|
| **All 19,140 questions** | **59.3** |
| Memory | 48.1 |
| Abstraction | 50.9 |
| Physics | 52.3 |
| Semantics | 77.0 |

Multiple-choice video question answering (three options per question, chance 33.3): top-1 accuracy (%), videos read with the defaults above.

Jev Decision Index: official leaderboard, text and vision boards 0.3.2, 2026-10-12; the boards evaluated v3.1.0, which has the same weights

as this version. Size bands and the Pareto frontier use the parameter counts listed by the boards. d3-lite on the public vision benchmarks and the Perception Test: internal evaluation.

## License

Apache-2.0 (LICENSE). Built on Qwen/Qwen3.5-0.8B (Apache-2.0).

## Citation

```
@misc{d3_lite_2026,
  title        = {{d3-lite}: A Multimodal Fou
  author       = {{vLLM Semantic Router Team}
               = {2026},
  howpublished = {\url{https://huggingface.co
}
```

Trained on AMD Instinct MI325X GPUs.

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